CABINET CHEAT SHEET

# Power Glove Vision Quick Reference

Current cabinet addresses, services, controls, and maintenance commands.

2026 EDITION / IAIN BENNETT / 

Use this reference with your own installation. Replace `UNO-Q-NAME.local` with your UNO Q hostname and `RETROPIE-NAME.local` with your RetroPie hostname. Commands run from your downloaded project directory unless stated otherwise. Never include private pairing tokens in documentation.

## UNO Q

| Item | Your installation |
|---|---|
| Board name | Your UNO Q board name |
| Network hostname | `UNO-Q-NAME.local` |
| IPv4 address | Check your router or the board network settings |
| Board | Arduino UNO Q |
| App Lab app | PowerGlove Vision |
| Camera | UVC-compatible USB camera, configured as `auto` |
| Receiver hostname | `RETROPIE-NAME.local` |
| Startup profile | Choose in Setup |

Prefer the `.local` hostname in bookmarks and configuration. The numeric IP is assigned by the network and may change.

## Browser URLs

| Purpose | URL |
|---|---|
| Live debug dashboard | <http://UNO-Q-NAME.local:8088/debug> |
| Offline gesture lessons | <http://UNO-Q-NAME.local:8088/learn> |
| Friendly connection setup | <http://UNO-Q-NAME.local:8088/setup> |
| Secure pairing setup | <https://UNO-Q-NAME.local:8443/setup> |
| Root (redirects to dashboard) | <http://UNO-Q-NAME.local:8088/> |
| Machine-readable status | <http://UNO-Q-NAME.local:8088/status> |
| Camera stream | <http://UNO-Q-NAME.local:8088/stream> |
| GitHub repository | <https://github.com/mathan416/PowerGlove-Vision> |

If `.local` discovery is unavailable, replace `UNO-Q-NAME.local` with the current UNO Q IP address shown by your router or board network settings.

## Network ports

| Port | Protocol | Direction | Use |
|---|---|---|---|
| `8088` | TCP | Browser -> UNO Q | Setup, dashboard, status and camera stream |
| `8443` | TCP/TLS | Browser -> UNO Q | Secure RetroPie pairing |
| `55355` | UDP | UNO Q -> RetroPie | Controller-state packets |
| `55356` | UDP | RetroPie -> UNO Q | Authenticated profile selection and acknowledgement |
| `55357` | TCP/TLS | UNO Q -> RetroPie | Temporary one-time pairing helper |

Keep these ports limited to the trusted local network. Do not expose them to the public internet.

## Service and configuration reference

| Item | Value |
|---|---|
| Console hostname | `RETROPIE-NAME.local` |
| Virtual controller | `PowerGlove Vision` |
| Pairing-token file | `/etc/powerglove/token` |
| Game registry | `/etc/powerglove/games.json` |
| Launcher settings | `/etc/powerglove/launcher.json` |
| Receiver service | `powerglove-receiver.service` |
| Delayed-start timer | `powerglove-receiver.timer` (45 seconds after boot) |
| UNO Q shutdown watcher | `powerglove-system-shutdown.path` (verify it is enabled and active) |
| Shutdown readiness marker | `data/.shutdown-enabled` |
| Shutdown marker boot rule | `/etc/tmpfiles.d/powerglove-system-shutdown.conf` |

The token in `/etc/powerglove/token` must exactly match the token stored in the UNO Q app's `data/device.json`. Never paste that value into GitHub, screenshots, logs or this cheat sheet.

Install the shutdown helper before using the shutdown control. **Stop controller** leaves the UNO Q running; **Shutdown** requests a Linux halt. The tested board restarts afterward, so remaining stopped is not guaranteed.

# Project files

| Purpose | Path |
|---|---|
| Git working copy | Your downloaded project directory |
| Installation guide | `docs/INSTALL_README.md` |
| Gameplay guide | `docs/GAMEPLAY_GUIDE.md` |
| Programs A-I reference | `docs/bad-street-brawler-programs.md` |
| UNO Q App Lab installation ZIP | `output/app-lab/PowerGlove-Vision-Uno-Q.zip` |
| Gesture profiles | `config/profiles.json` |
| Automatic game mapping | `config/games.json` |

# Deploy over Wi-Fi

Configure SSH access to your UNO Q first. From your local project directory, deploy application updates and restart PowerGlove Vision with:

```sh
scripts/deploy-uno-q-wifi.sh arduino@UNO-Q-NAME.local
```

One-time installation of the Dashboard and Setup shutdown control:

```sh
scripts/install-uno-q-shutdown-helper.sh arduino@UNO-Q-NAME.local
```

The helper asks for the UNO Q password through the terminal. **Shutdown** requests a graceful Linux halt. See the known restart limitation below.

This preserves the private `data/device.json`. No password is stored in the repository. USB is only needed as a recovery option or for a passwordless blue matrix firmware upload; App Lab can also perform a wireless firmware update by asking for the board password privately.

Verify key access before deploying:

```sh
ssh -o BatchMode=yes arduino@UNO-Q-NAME.local hostname
```

# Pair your RetroPie

Recommended one-time-code method:

1. On RetroPie, run `sudo /opt/powerglove/bin/powerglove-pair`.
2. Open <https://UNO-Q-NAME.local:8443/setup> on the trusted local network.
3. Enter `RETROPIE-NAME.local` and the 20-character code.

4. Prepare pairing and compare the matrix `ID` with the browser certificate's SHA-256 fingerprint prefix.
5. Enter the matrix `PN` digits and complete pairing.
6. Confirm `powerglove-receiver.service` is active.

Username/password pairing is available on the same secure page. Choose **Prepare password pairing** before entering the password, perform the same matrix certificate check, and then complete the one-time SSH operation. The password is not stored.

## Quick health checks

Open the status endpoint in a browser or run:

```
SH
curl -sS http://UNO-Q-NAME.local:8088/status
```

Healthy tracking should report:

```
JSON
{
  "camera_available": true,
  "worker_running": true,
  "detected": true,
  "calibrated": true
}
```

Useful RetroPie checks:

```
SH
sudo systemctl status powerglove-receiver.service
sudo systemctl status powerglove-receiver.timer
sudo journalctl -u powerglove-receiver.service -n 100 --no-pager
grep -A8 -B2 'PowerGlove Vision' /proc/bus/input/devices
```

Run the local software tests:

```
SH
PYTHONPATH=src python3 -m unittest discover -s tests -v
```

## Camera troubleshooting

Your UVC-compatible USB camera must be connected to the UNO Q through an externally powered USB-C hub. Connect to App Lab over Wi-Fi while the UNO Q is acting as the USB host. A camera connected to your computer is not visible inside the UNO Q app.

If the UNO Q lists only `qcom-venus-encoder` and `qcom-venus-decoder`, your camera is not visible to Linux. Check the powered hub, reconnect your camera, wait several seconds, and watch the dashboard. The supervisor retries automatically; do not cycle through camera indexes when no UVC device exists.

## Website images

- `docs/images/debug-dashboard.png`

- `docs/images/learn-page.png`
- `docs/images/setup-page.png`

# Profiles

Dashboard and Learn display **Starting camera and gesture tracking** with elapsed seconds during initialization. First startup may take longer than later profile switches. Wait for the camera view before centering. Gestures off leaves the camera off until a profile or Learn needs it.

The Dashboard can change the active profile for the current session, and Setup can choose the profile used at startup:

- Bad Street Brawler
- Super Glove Ball
- Power Glove Programs A-I
- Gestures off

RetroPie launch hooks select the registered profile for each recognized ROM. Launching an unregistered game safely selects **Gestures off**. Automatic profile selection currently applies only to NES and Famicom games; launching another system also turns gestures off.

## Useful off-script starting points

| Program | Good first experiment | Controls and tradeoff |
|---|---|---|
| A | Pinball and games built around short bursts | Finger actions suit flippers, wrist movement can suit tilt, and pullback toggles a control. It does not provide an ordinary D-pad. |
| D | Challenge runs and games where reversed movement is fun | Every direction is reversed; thumb and index actions provide A and B. |
| H | General NES and Famicom experiments | Hand movement provides a conventional D-pad and thumb/index actions pulse A and B. Pulses are a poor fit for actions that must be held continuously. |

## Quick off-script test

1. Launch an unregistered NES or Famicom game. Confirm that PowerGlove Vision reports **Gestures off**.
2. Open the UNO Q **Dashboard** and choose **A: Pinball**, **D: Challenge**, **H: General**, or another profile.
3. Wait for vision to start using your saved calibration.
4. Recalibrate only if neutral needs adjusting, select **Start controller**, and test movement and actions.
5. Stop the controller before changing profiles or games.

This temporary choice is ideal for discovery. It does not change the automatic game registry or the startup profile saved in Setup. While **Gestures off** is selected, the camera and MediaPipe tracker are closed and the UNO Q matrix runs its animated glove attract sequence.

### Keep a working combination

Add the ROM's exact basename, including `.nes`, `.zip`, or `.7z`, to the `games` object in `/etc/powerglove/games.json`. Preserve all existing entries. For example:

```
JSON
{
  "games": {
    "YOUR EXACT GAME FILENAME.nes": "program_h"
  }
}
```

Check the edited file before testing:

```
SH
sudo python3 -m json.tool /etc/powerglove/games.json >/dev/null
```

Restart the game so its launch hook selects the newly registered profile. See `docs/GAMEPLAY_GUIDE.md` for the illustrated play and experimentation guide and `docs/bad-street-brawler-programs.md` for complete Program A-I controls.

For Super Glove Ball, hand position controls the D-pad, index curl is A, thumb curl is B, a V sign held for about 0.7 seconds is Start, and a thumbs-up with the other fingers closed is Select.

## Startup behaviour

In Arduino App Lab, the current PowerGlove Vision project must show the `DEFAULT` badge. The app then starts automatically after the UNO Q boots. The dashboard becomes available before the camera worker is ready, which makes it the best first place to diagnose startup or camera problems.

On RetroPie, direct boot enablement of `powerglove-receiver.service` is disabled. `powerglove-receiver.timer` starts it after 45 seconds so EmulationStation initializes before the virtual controller appears. This helps prevent frontend pauses and conflicts with other USB devices, including the BitPixel display, when the receiver starts too early.

App Lab should contain one active `powerglove-vision` installation. Keep that copy set as the default/autostart app so the dashboard returns after every UNO Q reboot. The Wi-Fi deployment script reapplies this designation.

## Known limitation: UNO Q restarts after Shutdown

On the tested UNO Q, a graceful `halt` still leads to an automatic restart, including when connected directly to a Mac without the powered hub. The helper requests halt correctly, but remaining stopped is not verified. Do not use loss of the website or a fixed delay as confirmation that power can safely be removed. See the installation guide for the investigation status and Arduino guidance.

## Saved calibration

Calibrate once with a relaxed hand in your neutral playing position. The reference in `data/calibration.json` survives Learn, profile changes, and restarts. Recalibrate when the camera or your position changes, or neutral is wrong. Include this file in private backups. Camera-overlay Right/Left and its confidence score describe handedness, not movement.